PALANTIR FOUNDRY PROJECT ENABLEMENT PROGRAM

Phase 1 — Day 1

Project Architecture & Solution Walkthrough

2 Hours • 10-Day Foundation Accelerator

Presented by Rajesh Pasham

PROGRAM RECAP

Where Today Fits In

Phase 1

Project Technology Accelerator

10 Days × 2 Hours

→

Phase 2

Guided Project Development

5 Weekly Sprints

→

Phase 3

Office Hours & Mentoring

YOU ARE HERE

Day 1 of Phase 1 sets the shared architectural foundation the whole engagement builds on.

SESSION OBJECTIVES

By the End of Today, You Will Be Able To

- Describe the major architectural layers of the Foundry platform
- Explain the Ontology's core building blocks: object types, properties, link types, and action types
- Navigate the Ontology Manager and locate key resources
- Describe how Workshop, OSDK applications, and Functions consume the Ontology
- Create your first Ontology object types in your own Developer Tier account
- Understand the coding standards and sprint structure for the rest of the engagement

PLATFORM ARCHITECTURE

The Foundry Platform, Built Bottom-Up

Applications	Workshop dashboards and OSDK-powered apps (React, Python, Java)
Functions & Automation	Business logic, event triggers, scheduled jobs
Ontology	Object types, properties, link types & action types — the semantic layer
Data Integration	Connects source systems and files into Foundry datasets

Each layer is built on top of the one below it — start with data, then model it, then automate it, then expose it.

CORE CONCEPTS

What Is the Ontology?

- A categorization of your organization's world — Palantir describes it as a digital twin
- Maps real datasets and models onto object types, properties, link types, and action types
- Object Type

Defines an entity or event — e.g. Vehicle, Transponder

- Property

A characteristic of an object type — e.g. status, make, year

- Link Type

A relationship between two object types — e.g. Vehicle ↔ Transponder

- Action Type

Defines how an object can be created or modified

A Familiar Analogy

A Dataset...

- A dataset is like a spreadsheet
- A row is one record
- A column is one field

...and an Object Type

- An object type is like a dataset
- An object is one instance (one row)
- A property is like one column

WORKED EXAMPLE

Object Type: Vehicle

Property	Type	Notes
vehicleId	String	Primary key / title
vin	String	17-character VIN
make / model	String	e.g. Ford, Transit-350
year	Integer	
licensePlate	String	
status	String	Active / Maintenance / Inactive
currentLocation	String	Facility name

Link Type: Vehicle ↔ Transponder

- Cardinality: one-to-one for Day 1 (a vehicle has at most one active transponder)
- Foreign key: Transponder.assignedVehicleId → Vehicle.vehicleId
- Later sprints extend this to one-to-many for full lifecycle history

CORE CONCEPTS

Action Types — A Quick Preview

- Action types define how an object can be created, updated, or deleted
- Every write to the Ontology goes through an action — never a direct database write
- Actions can be triggered from Workshop buttons, APIs, or Functions

We build our first custom action types in Week 2 (Advanced Ontology Modeling)

APPLICATIONS

Workshop — Operational Applications on the Ontology

- Workshop is Foundry's no-code / low-code application builder
- Widgets (tables, forms, maps, charts) bind directly to Ontology objects and actions
- Used for internal, operational tools — dashboards, review queues, data entry

In this engagement: Inventory, Shipment, and Check-in/Check-out screens are built in Workshop

DEVELOPER TOOLING

OSDK, Developer Console & Foundry CLI

- Ontology SDK (OSDK)

A generated, typed client library (TypeScript, Python, Java) for reading and writing Ontology objects from code

- Developer Console

Where OSDK applications and OAuth clients are created and managed

- Foundry CLI & Code Repositories

Local development, version control, and publishing pipelines for Functions and applications

- SuperRepo (newest capability)

Define Ontology, Functions, and frontend together in one repository, with the OSDK generated automatically

LOOKING AHEAD

React + OSDK Architecture Preview

React PWA → OSDK Client (generated) → Ontology (Objects & Actions)

Modules 2–3 (Days 3–6) build this end to end, starting with your first OSDK application on Day 2.

Coding Standards for This Engagement

- Object & property naming: lowerCamelCase for properties, PascalCase for object type display names
- TypeScript for all React/OSDK application code; Python for Functions and data transforms
- One Code Repository per application; Ontology changes reviewed before publishing
- Every Pull Request references its Sprint and includes a short description of the Ontology or UI change
- Commit early, commit often — small, reviewable changes over large batch commits

LOOKING AHEAD

Phase 2 Sprint Plan — Recap

Sprint	Focus
Sprint 1	Ontology — Shipment, Inventory, Transponder, Bin
Sprint 2	React App — Add / Remove Transponder
Sprint 3	Shipment APIs — External integrations, polling
Sprint 4	Barcode, Inventory & Geospatial Visualization
Sprint 5	Testing, Deployment, Security

Everything you build in Phase 1 becomes the foundation for these sprints.

HANDS-ON, STARTING NOW

What You'll Build in the Next 40 Minutes

- Sign in to your own Foundry Developer Tier account
- Create the Vehicle object type with 8 properties
- Create the Transponder object type with 6 properties
- Link them with a one-to-one relationship
- Upload the provided sample datasets and back your object types with real data

Full step-by-step instructions are in today's Hands-On Exercise document

Questions?

Open floor — architecture, tooling, or anything from today's walkthrough.

THANK YOU

Next Session — Day 2

Module 1: OSDK Architecture

- OSDK overview and SDK concepts
- Architecture and Object APIs
- Actions and Queries
- Authentication
- *Hands-on lab: develop your first OSDK application*

Rajesh Pasham